

Joseph A. Falcone

(917) 935-5075 | jfalcon4@vols.utk.edu | linkedin.com/in/falconejoseph

PROFESSIONAL SUMMARY

M.S. Business Analytics and AI candidate with experience in data engineering, machine learning, analytics, relational database design, and optimization. Skilled in Python, R, SQL, and Tableau, with experience building end-to-end data workflows, improving data quality, and transforming complex datasets into analytical and decision-support solutions.

EDUCATION

University of Tennessee, Haslam College of Business

Knoxville, TN

Master of Science in Business Analytics and AI

Expected May 2027

Bachelor of Science in Business Administration

May 2026

TECHNICAL SKILLS

Data Science & Machine Learning: Pandas, NumPy, scikit-learn, XGBoost, TensorFlow, predictive modeling, model evaluation, statistical analysis

Data Engineering: relational database design, data cleaning, data transformation, data integration, data quality

Optimization: Pyomo, mathematical programming, constrained optimization

Visualization: Tableau, Matplotlib, ggplot2

EXPERIENCE

HASLAM COLLEGE OF BUSINESS, *Data Support Specialist*, Knoxville, TN

Aug 2026 – Present

- Built an integrated alumni and employer engagement data framework by transforming fragmented records into a structured foundation for analysis and reporting.
- Designed relational database models and workflows to improve data quality and support reliable tracking.
- Built analytics and reporting solutions that converted raw engagement data into actionable information for Graduate Business Programs alumni and career-development teams.

NYPD HEADQUARTERS, *Business Process Improvement Aid*, New York, NY

Jun 2025 – Jan 2026

- Collaborated across Personnel departments to align project requirements and ensure accurate implementation of process improvement strategies.
- Developed dashboards, KPIs, process documentation, and reporting materials to improve clarity, communication, and understanding across teams and stakeholders.

SELECTED TECHNICAL PROJECTS

NEURAL NETWORK BUILD – TensorFlow, Neural Networks

Sep 2026

- Built a three-hidden-layer feed-forward neural network to predict product quantity sold from large e-commerce data using incremental mini-batch learning.
- Evaluated performance with train/test R^2 and moving-average MSE while tracking RAM usage, training time, variable importance, and partial dependence.

COLLEGE BASKETBALL ROSTER OPTIMIZATION – XGBoost, Predictive Modeling & Optimization

Sep 2026

- Developed an XGBoost model to project player performance and identify undervalued roster options relative to asking price.
- Combined predictive modeling with constrained optimization to maximize projected performance under an \$8 million budget; won first place in the graduate division.